

Surrogate-Assisted Genetic Optimization of a PSA Process

Manifest-only CO₂ capture optimization with NSGA-II

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SEEN5320 Final Project

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Goal

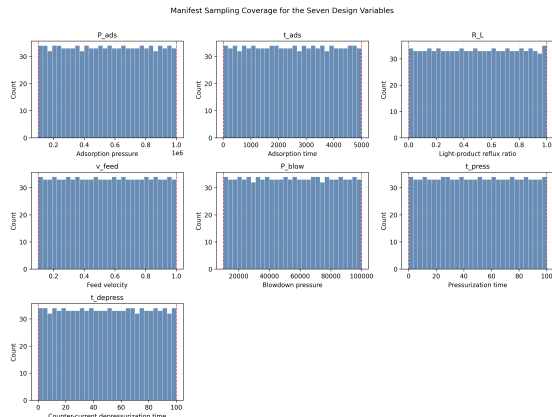
Build a closed surrogate-assisted optimization workflow using only `data/manifest.csv`.

- Removed dependency: deleted time-space CSV files and derived features.
- Surrogate inputs: seven PSA operating variables x_1 – x_7 .
- Surrogate outputs: purity, recovery, productivity, and specific energy.
- Optimizer: NSGA-II genetic algorithm coupled to the surrogate.
- Closed loop: export optimized inputs for detailed-model reruns and compare returned outputs.

Manifest Dataset

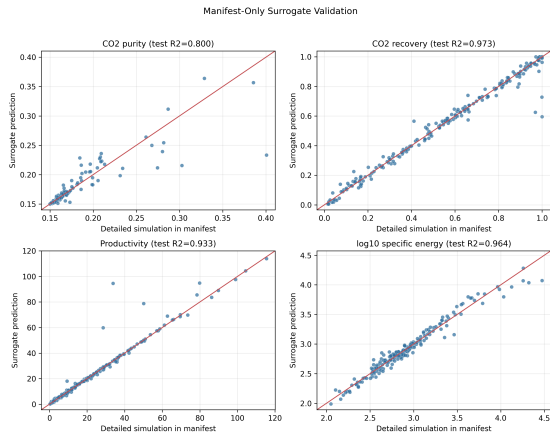
Item	Value
Successful simulations	1000
Design variables	7
KPI outputs	4
Derived time-space features	0
Train/test split	80/20

- Bounds are read from `ProcessConfig.yaml`.
- Energy is modeled in log space because it is highly skewed.

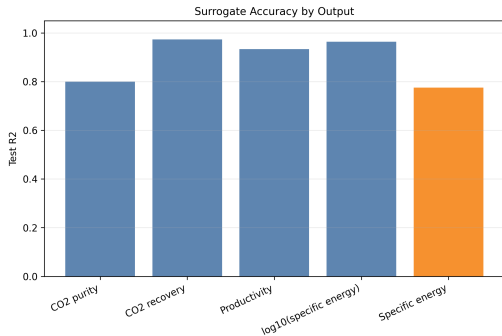


Surrogate Model

- Degree-3 ridge response surface in standardized design variables.
- Purity target:
 $\log_{10}(\text{purity} - 0.15)$.
- Other targets: recovery, productivity, and $\log_{10}(\text{energy})$.
- Ridge regularization selected by five-fold cross-validation.
- Output predictions clipped to the observed manifest range before GA search.



Surrogate Accuracy



Target	RMSE	Test R^2
Purity	0.0166	0.800
Recovery	0.0502	0.973
Productivity	5.50	0.933
$\log_{10}(\text{energy})$	0.0817	0.964

- Purity is the hardest scalar-output target.
- Recovery, productivity, and log-energy are reliable enough for screening.

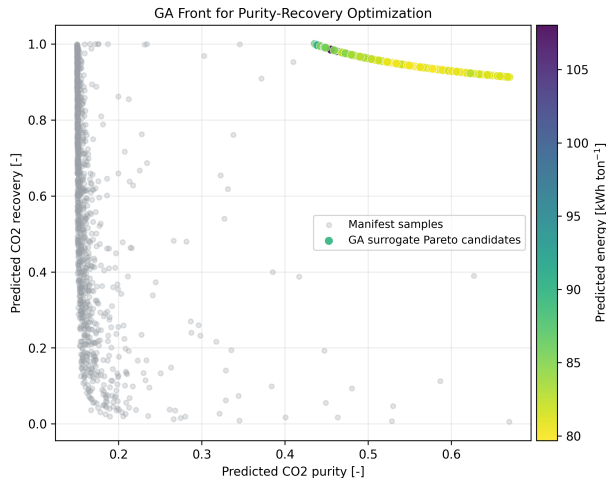
NSGA-II settings

- Population size: 120.
- Generations: 200.
- Simulated binary crossover: 0.9.
- Polynomial mutation: $1/7$ per variable.
- Initial population: manifest-seeded plus random points.

Two bi-objective fronts

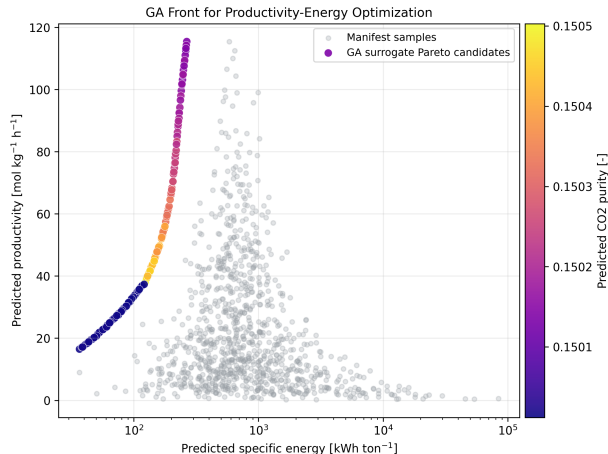
- Maximize purity and recovery.
- Maximize productivity and minimize energy.
- Export 20 representative candidates per front for detailed-model reruns.

Front 1: Purity vs Recovery



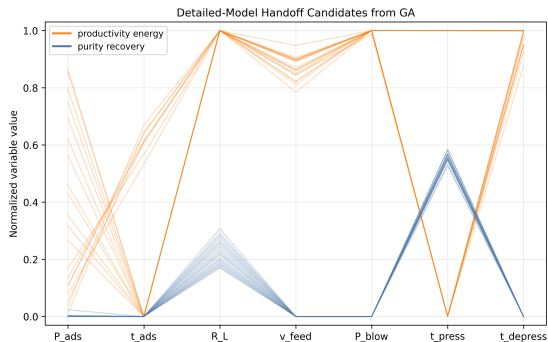
- Predicted purity range: 0.436–0.670.
- Predicted recovery range: 0.913–1.000.
- Productivity remains low: about 5.5–5.9 mol kg⁻¹ h⁻¹.
- These candidates are separation-quality screening points.

Front 2: Productivity vs Energy



- Predicted productivity range: 16.49–115.45 mol kg⁻¹ h⁻¹.
- Predicted energy range: 36.70–266.08 kWh ton⁻¹.
- Purity remains near feed level.
- This front is mainly a throughput-energy tradeoff.

Closed-Loop Handoff



- Full final GA populations are saved as CSV.
- Surrogate Pareto candidates are saved separately.
- Forty rerun inputs are exported for the detailed model.
- A result template defines the expected return format.
- The comparison script computes errors after detailed results are saved.

- The project now uses only existing manifest data for surrogate training.
- NSGA-II turns the surrogate into two explicit multi-objective design fronts.
- Output clipping prevents the GA from reporting values outside the manifest training envelope.
- Purity-optimized candidates require detailed-model confirmation because purity is the weakest surrogate target.
- The CSV handoff completes the closed-loop structure for surrogate-versus-detailed comparison.